Investigating CTT in LSTMs: Kill/Birth Ratios

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Abstract

Complementary Turnover Training (CTT) is a biologically-inspired paradigm in which neurons are permanently killed and replaced during training, drawing on adult neurogenesis and programmed cell death. Existing work fixes turnover to a small symmetric rate and validates it on supervised sequence tasks, leaving the balance between destruction and rebuilding untested as an experimental variable. We isolate that balance: we augment an LSTM policy with a plastic turnover layer, train it by REINFORCE on a partially-observable, egocentric maze-navigation task, and sweep eleven kill/birth ratio conditions spanning net-shrinking, balanced, and net-growth regimes, together with a death-only ablation and three regularised baselines, holding the parameter count identical so that only the ratio varies. All fourteen conditions are trained over eight seeds and evaluated zero-shot on progressively larger mazes. At low seed counts, and on the largest mazes, higher-turnover and net-growth regimes initially appeared to generalise better, but this effect does not survive statistical analysis. Two effects, however, are real: death-only turnover is worst on every axis (-17 pp at 13×13), and CTT carries a trained-in path-efficiency cost that widens with maze size. We conclude that the task and environment chosen was not memory-intense enough to benefit from CTT, and that the kill/birth ratio behaves as a capacity knob rather than a representation-quality knob: its effect on generalisation is bounded near zero, and CTT's benefits, consistent with its damage-tolerance framing, are robustness rather than generalisation.

Keywords

neural turnover, structural plasticity, adult neurogenesis, recurrent neural networks, reinforcement learning, generalisation, damage tolerance

1 Introduction

Artificial neural networks must reconcile two competing demands. They must be stable enough to retain what they have already learned, but plastic enough to improve as training proceeds. This is usually managed implicitly through the optimiser and regulariser rather than confronted directly. In recurrent policies trained by reinforcement, where the learning signal is sparse and the same weights must encode both an implicit world model and a control policy over long horizons, the problem is even more apparent: capacity committed early to a mediocre strategy is difficult to fix.

The dominant tools for plasticity in recurrent networks, dropout and zoneout, share a structural feature: they perturb the network temporarily. Dropout masks a random subset of units on each forward pass; zoneout instead randomly preserves a unit's previous hidden state, injecting stochasticity into the recurrence rather than the activations. However, since no unit is ever removed, and no capacity renewed, the network ends training with exactly the neurons it began with, so these methods can regularise how existing capacity is used, but they cannot change what capacity exists.

Biological neural systems, however, are not restricted in this way. In the adult human brain, the dentate gyrus continues to generate new neurons throughout life (adult neurogenesis), and these new neurons integrate into circuits that support learning and memory [2], while programmed cell death (apoptosis) removes others and homeostatic mechanisms rescale activity to keep the network stable [17], so that the population is continually turned over rather than merely reused. Complementary Learning Systems (CLS) theory [12] frames this at the systems level as a division of labour between fast, plastic and slow, stable subsystems. Complementary Turnover Training (CTT) imports the idea into gradient-based learning: during training, neurons in a designated plastic zone are permanently killed and replaced with freshly initialised units, on a fixed schedule, so that the network's capacity is genuinely renewed rather than only ever re-weighted.

Existing turnover work treats turnover as a binary intervention, and validates it almost exclusively on supervised classification benchmarks, and the balance between destruction and rebuilding has never been isolated as an experimental variable. The canonical CTT formulation fixes a small, symmetric turnover rate, leaving open whether it is turnover per se that matters or the specific ratio of kills to births. A net-shrinking schedule (more kills than births) permanently contracts capacity, a net-growth schedule (more births than kills) expands into reserve capacity, and a balanced schedule holds it constant. These are qualitatively different interventions that the literature aims to test.

1.1 Research Question

This study addresses a single research question: within Complementary Turnover Training, does the kill/birth ratio (the ratio of plastic-layer units killed to plastic-layer units rebirthed at each turnover event) govern how well a CTT-based recurrent policy generalises zero-shot to mazes larger than any seen in training? We augment an LSTM policy with a plastic turnover layer, train it by REINFORCE on a partially observable, egocentric maze-navigation task, and sweep eleven kill/birth ratio conditions spanning the net-shrinking, balanced, and net-growth regimes, together with a death-only ablation in which killed units are never replaced. Because the ratio is the single design dimension the source method leaves fixed, isolating it determines whether it is turnover, or a specific balance of destruction and rebuilding, that has any effect on generalisation.

The **independent variable** is the kill/birth ratio, which we decompose into two axes: turnover magnitude (the total fraction of units replaced per event) and regime (the net direction of change,

shrinking through balanced to growing); the architecture, the parameter count, and every other element of the turnover cycle are held fixed, so the ratio is the sole manipulated variable. The **dependent measures** are zero-shot solve rate and path suboptimality (agent steps relative to the breadth-first-search optimum) evaluated on mazes larger than the training size. The **scope** is deliberately narrow: a single family of perfect mazes, trained at 7×7 and evaluated without further training at sizes from 5×5 to 13×13. Before running any evaluation, we predicted that balanced turnover would generalise better than unbalanced turnover; we find no such effect. There were three ways this could have resolved: the prediction could hold, with the balanced conditions dominating at the extrapolation sizes; a directional but unpredicted effect could emerge, with magnitude or regime predicting generalisation along one of the two axes; or no regime could separate from the baselines.

2 Related Work

This section reviews the literatures that inform our design and identifies where they leave the kill/birth ratio untested. Section 2.1 covers the stochastic regularisers that furnish our baselines; Section 2.2 covers structural, turnover-inspired alternatives to stochastic regularisation, from which CTT descends; Section 2.3 details CTT itself, the method this study modifies; Section 2.4 covers the separate literature on scale generalisation, the phenomenon we measure; and Section 2.5 draws these together into the specific gap this study addresses.

2.1 Regularisation and Plasticity in RNNs

Regularisation of recurrent networks has largely developed as a family of stochastic disruption methods. Dropout [15] randomly zeros a fraction of activations during training to discourage co-adaptation; applied naively to the recurrent connections of an RNN it damages long-range memory, which motivated variants that fix a single mask across time steps (variational dropout [5]) and that adapt the dropout rate during training (fast and adaptive dropout). Zoneout [11] takes a complementary approach tailored to recurrence: rather than zeroing activations it stochastically retains a unit's hidden state from the previous step, preserving gradient flow while still injecting noise. We adopt vanilla, dropout, and zoneout LSTMs as baselines because they are the strongest and most widely used members of this family for recurrent policies, and because they share our backbone exactly, isolating the regulariser as the only difference. What unifies these methods is that they neither select units by importance nor free capacity.

2.2 Neurogenesis and Turnover-Inspired Learning

A smaller line of work takes structural change, rather than stochastic masking, as its mechanism. Connection pruning removes low-salience weights to compress a trained network [6], establishing that a network can shed capacity without a catastrophic loss of function. Renewal is the complementary idea: Neurogenesis Deep Learning [4] grows an autoencoder by adding units as new data arrives, and approaches framed around adult neurogenesis as a neural regulariser [16] periodically introduce or reinitialise units to maintain plasticity. The importance of such renewal is highlighted by recent evidence that conventional deep networks progressively *lose* plasticity under sustained training unless variability is actively reinjected [3], and that units in deep reinforcement-learning agents can fall dormant and are productively reset [14]. This structural line is complementary to weight-based continual-learning methods, which instead mitigate catastrophic forgetting by constraining the parameters important to earlier tasks [10]. These methods establish that adding, resetting, or renewing units can help, yet they share two limitations relative to the question we pose: none treats the *balance* of destruction and rebuilding as a tunable variable, and with the exception of dormant-unit resetting in reinforcement learning [14], they are evaluated almost exclusively on supervised classification, where credit assignment is immediate and the parameters do not additionally have to encode a control policy.

2.3 Complementary Turnover Training

Complementary Turnover Training (CTT), introduced in the source work on which this study builds, formalises neuron turnover as a repeating five-stage cycle applied to a designated plastic zone during training. At each turnover event the cycle scores every plastic unit by a combined criterion that rewards importance and penalises redundancy; kills the lowest-scoring fraction, subject to the protections described in Section 3.3; births replacement units by cloning a high-importance donor and adding Gaussian noise (clone-mutate); gates each newborn in gradually, ramping its output gain from zero over a fixed window so it does not disrupt a working policy; and restores homeostasis and consolidates through a short replay step. The method has four principal design dimensions: the scoring rule, the rebuilding strategy, the stabilisation machinery, and the capacity-rebuilding ratio (the balance of kills to births). We vary only ratio and inherit the other three unchanged from the source configuration, so our results speak directly to the balance question the source work left fixed.

The benefit that CTT currently validates is damage tolerance on supervised sequence-memory tasks, which is explicitly regime-conditional: it is largest where memory demand is highest and vanishes on tasks where all methods already saturate, and secondly, the source work does not claim to solve catastrophic forgetting.

2.4 Scale Generalisation in Navigation

A separate literature asks whether recurrent and iterative networks can extrapolate to problem instances larger than those seen in training. Work on maze navigation and algorithmic reasoning has shown that recurrent architectures can, under some conditions, extrapolate from easy to hard instances by iterating a learned update for more steps at test time [13]. This literature studies the maze rather than architecture-side mechanisms of plasticity, and so says little about whether a turnover schedule imposed during training helps or hinders extrapolation, which is the gap this paper addresses.

Regarding architecture, we use an LSTM rather than a simpler recurrent unit or a feedforward network because the task is partially observable: the egocentric agent sees only a local window and a goal bearing, so solving a maze requires integrating observations over many steps and remembering where it has already been. Feedforward and reactive policies cannot represent this history, and simpler recurrent units train poorly over the long horizons the

larger mazes demand. The LSTM's gated memory [7] is the minimal architecture that makes the task learnable while remaining small enough to sweep 112 training runs, and its single recurrent layer gives the turnover zone a well-defined place to act.

2.5 Summary: The Gap This Study Addresses

Across these three literatures, a consistent gap emerges. Regularisation methods (Section 2.1) perturb existing capacity but never renew it; renewal methods (Section 2.2) show that adding, resetting, or renewing units can help, but none treats the *balance* of destruction and rebuilding as a tunable variable, and nearly all are validated on supervised classification rather than reinforcement learning. CTT itself (Section 2.3) fixes this balance at a single, small, symmetric rate, and its one validated benefit — damage tolerance on supervised sequence memory — is regime-conditional rather than general. The scale-generalisation literature (Section 2.4), meanwhile, studies extrapolation without addressing whether a turnover schedule helps or hinders it. No existing work isolates the kill/birth ratio as an experimental variable on a task where extrapolation is the outcome of interest and credit assignment is not immediate, which motivates the design in Section 3.

3 Methodology

This section describes how the research question (Section 1.1) is answered by our experiments.[1] Section 3.1 describes the task and environment; Section 3.2 the agent architecture and baselines; Section 3.3 the turnover cycle; Section 3.4 the experimental conditions; Sections 3.5 and 3.6 the training and evaluation protocols; and Section 3.7 implementation and reproducibility.

3.1 Task and Environment

Mazes are procedurally generated perfect mazes: a maze of size n is a spanning tree over an n × n cell grid, embedded in a (2n + 1) × (2n + 1) pixel grid whose walls separate the cells, with exactly one path between any two cells (no loops). The start is fixed at the top-left corner and the single goal at the bottom-right corner. Perfect mazes make the shortest path well defined and let us score every rollout against a breadth-first-search (BFS) optimum.

At each step, the agent observes a 5 × 5 wall patch centred on its current cell (window radius 2), with out-of-bounds cells treated as walls; a relative-goal vector from agent to goal, normalised by a fixed constant; and a one-hot encoding of its previous action. Concatenated, this is a 31-dimensional observation whose size is independent of the maze size. This is the central design choice that makes zero-shot transfer possible: because local wall patterns and the goal bearing obey the same rules at every scale, a single policy trained at one size can be run, with no change to its input shape, on a maze of any other size. It departs deliberately from the full-grid observation described in the original proposal, which would have tied the input dimension to the maze size and made cross-size evaluation impossible; the egocentric encoding trades global information for size-invariance, which is exactly what the research question requires.

[1] Code, trained-model configurations, and full results for every experiment reported in this paper are available in this repository: https://github.com/ndelfava/BIONNHONSProject.

The reward is potential-shaped by BFS distance to the goal, which densifies an otherwise sparse signal. On a normal move the agent receives the reduction in its BFS distance to the goal minus a per-step cost of 0.05, so moving one cell closer yields a small net-positive reward and moving away a net-negative one; attempting to move into a wall costs 0.1 and does not change position; reaching the goal yields a terminal reward of 20. Episodes are capped at 400 steps.

Training, validation, and evaluation mazes are drawn from disjoint generator-seed ranges under a strict convention that prevents any held-out maze from appearing in training. Each seed's training bank is 500 mazes drawn from generator seeds at or above 10 000; validation uses a reserved band of 100 mazes; the zero-shot evaluations of Section 3.6 use their own separate high-numbered bands. Because the generator is deterministic in its seed, every condition and seed is evaluated on identical maze banks, making all cross-condition comparisons exactly paired.

3.2 Agent Architecture and Baselines

All fourteen conditions share an identical backbone, so any measured difference is attributable to the intervention and not to capacity or architecture. Observations pass through a small multilayer-perceptron encoder into a single LSTM cell with a 256-unit hidden state, whose output feeds two linear heads: a policy head over the four movement actions and a scalar value head for the REINFORCE baseline. The three baselines instantiate this backbone directly: vanilla applies no perturbation; dropout applies dropout to the recurrent output; zoneout stochastically retains hidden-state components across steps. The CTT agent inserts a plastic turnover layer on the LSTM's hidden state, and both heads read from the plastic layer's output; the turnover cycle acts only on this layer, leaving the LSTM and heads structurally untouched. To separate the effect of turnover from the mere presence of this extra layer, we additionally train a *static-plastic* control: the identical plastic layer with the turnover cycle disabled, so that all 256 active units persist and are trained by gradient descent alone—no unit is ever scored, killed, or reborn. The control is trained and evaluated under exactly the same protocol as every other condition, and lets us attribute the path-efficiency cost of Section 4.5 to turnover itself rather than to the plastic layer.

3.3 The Turnover Cycle

Every plastic unit carries a combined health score, importance $- \beta \cdot$ redundancy. Importance is the product of two exponential moving averages (decay 0.99) so a unit counts as important if it is both active and influential on the loss. Redundancy is the unit's maximum absolute correlation with any other plastic unit, penalising units that merely duplicate an existing one ($\beta = 0.5$). Units with the lowest combined score are the death candidates.

Three rules keep scoring from destabilising a working policy: the top 10% of units by importance are immune from death; newborn units are immune until they have matured (1 000 updates); and a minimum-alive floor guarantees that at least 70% of the initially active units stay alive at all times. Together these bound how much of the representation any single turnover event can disturb.

Replacement units are not random. Each newborn clones the weights of a donor sampled from the top 20% of units by importance

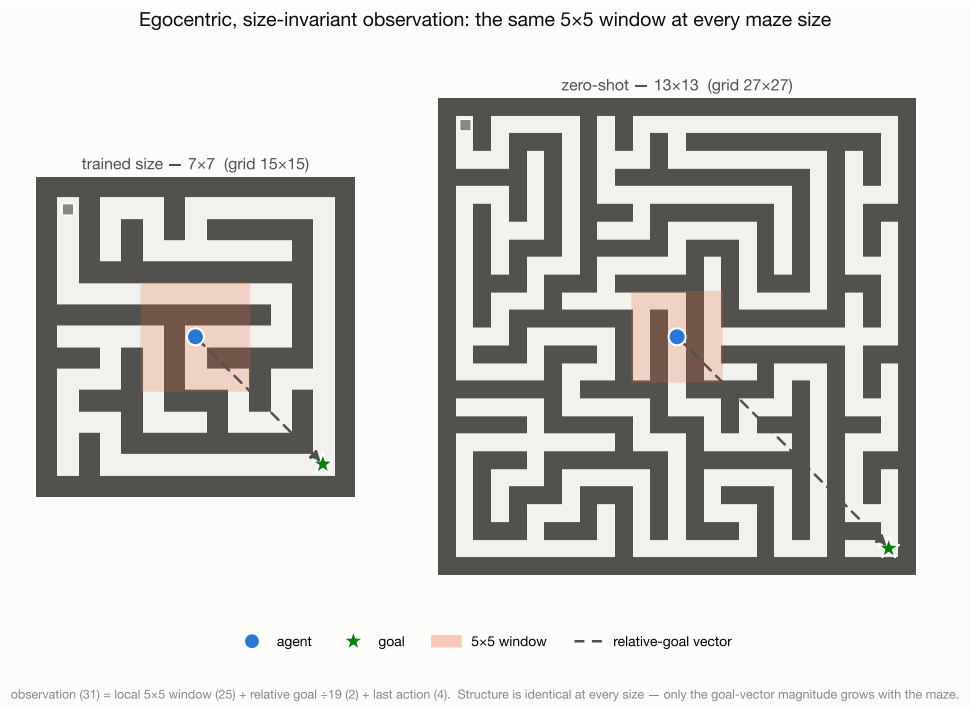


Figure 1: The egocentric observation. Both panels show the identical 5×5 window (orange) centred on the agent (blue), the goal (green star), and the relative-goal vector (dashed arrow) connecting them; the only difference between the trained 7×7 maze (left) and the zero-shot 13×13 maze (right) is the goal-vector magnitude, since the window itself never changes size. This is what makes a single policy, with a fixed 31-dimensional input, transferable to any maze size without retraining.

and adds small Gaussian noise ($\sigma = 0.02$), so a newborn starts near a useful function and then differentiates. Newborns enter gated off, and their output gain ramps linearly from zero to one over 1 000 updates, so their influence grows only as they mature.

After each turnover event, an activation-homeostasis step rescales unit gains toward a target activity level, and a short replay-consolidation step takes a few behaviour-cloning gradient steps against a replay buffer, damping the disruption of having just swapped units. The plastic zone has a fixed capacity of 384 slots, of which 256 are active at initialisation and the remaining 128 form a dormant reserve. A net-growth condition (births > kills) expands into this reserve; a net-shrinking condition (kills > births) contracts below 256; a balanced condition holds the active count constant. Because the slot count is identical in every condition, growth and shrinkage cost nothing in model size, and the kill/birth ratio is genuinely the only thing that differs across the eleven CTT conditions. This is the design decision that makes the research question answerable.

Two issues surfaced during validation and are worth recording because they affect correctness. First, the initial population must be granted maturation immunity: without it, the first turnover event can cull freshly-seeded units before their importance statistics have stabilised. Second, homeostasis must be applied at evaluation as well as training; the homeostatic gains are part of the learned function, and restoring the trained weights without them yields a different, degraded policy. Our checkpoint round-trip restores the homeostasis state explicitly.

The source method was tuned for supervised tasks running for many thousands of updates; its gate ramp, for instance, spans 2000 updates. Our RL agents converge in roughly 1500–2000 updates, so we rescaled the schedule proportionally: turnover runs every 500 updates, begins only after a 1000-update warmup during which a policy can form, and uses a 1000-update gate ramp and maturation window. These are the only schedule changes; the mechanism itself is unaltered.

3.4 Experimental Conditions

Table 1 lists the fourteen conditions. Eleven are CTT ratios, each defined by a (kill, birth) pair and arranged so that turnover magnitude (kill + birth) and regime (the sign of birth − kill) vary independently: at each of three magnitude levels we include a shrinking, a balanced, and a growth condition, and a death-only condition (kill without any birth) anchors the zero-rebuilding extreme. The remaining three are the regularised baselines. Every CTT hyperparameter other than the (kill, birth) pair is held fixed at the values in Section 3.3, so the design isolates the ratio.

3.5 Training Protocol

Agents are trained with REINFORCE [18] using a learned value baseline, entropy annealing, gradient clipping, and a step-decayed learning rate (Table 2). We use Adam [9] at a 5×10^{-4} learning rate, discount $\gamma = 0.99$, value-loss weight 0.25, a batch of 16 episodes per

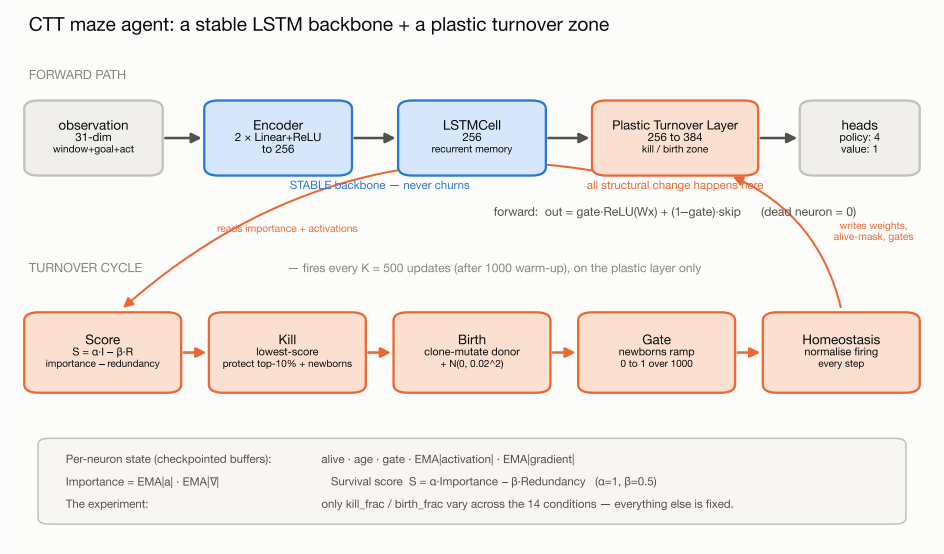


Figure 2: Agent architecture and the turnover cycle. Top row: the forward path, encoder through LSTM (blue, the stable backbone that never turns over) to the plastic turnover layer (orange, where all structural change happens) and the policy/value heads. Bottom row: the five-stage turnover cycle (score, kill, birth, gate, homeostasis) that acts on the plastic layer only, every 500 updates after a 1000-update warm-up.

Table 1: The eleven CTT kill/birth ratios (magnitude × regime), the three regularised baselines, and the static-plastic control (a plastic layer with turnover disabled). Kill and birth are fractions of the alive population per turnover event.

Condition	Kill %	Birth %	Ratio (k:b)	Regime
death_only	5	0	1:0	death-only
unbalanced_low	5	2	5:2	shrinking
balanced_low	5	5	1:1	balanced
net_growth_low	5	10	1:2	growing
unbalanced_mid	10	5	2:1	shrinking
balanced_mid	10	10	1:1	balanced
net_growth_mid	10	20	1:2	growing
unbalanced_high	20	10	2:1	shrinking
balanced_high	20	20	1:1	balanced
net_growth_high	20	30	2:3	growing
high_balanced	30	30	1:1	balanced
vanilla (baseline)	—	—	—	baseline
dropout (baseline)	—	—	—	baseline
zoneout (baseline)	—	—	—	baseline
static_plastic (control)	0	0	—	no turnover

update, gradient-norm clipping at 1.0, and an entropy coefficient annealed from 0.08 to 0.03. Training is capped at 4 000 updates with early stopping (patience 2000). Every agent is trained on 7 × 7 mazes only; the larger sizes used for evaluation are never seen in training. The full design is 8 seeds × 14 conditions = 112 agents. CTT adds roughly a 1.5× per-update overhead from the scoring and consolidation steps.

Checkpoints are selected by a rule identical across all conditions: highest validation solve rate, ties broken by fewest mean steps. This avoids saving an early checkpoint once solve rate saturates at ceiling, when only path efficiency still distinguishes checkpoints.

Table 2: Training hyperparameters, identical across all fourteen conditions.

Hyperparameter	Value
Optimiser	Adam
Learning rate	5×10^{-4}, StepLR (×0.9)
Discount γ	0.99
Value-loss weight	0.25
Entropy coefficient	0.08 → 0.03 (annealed over 16 000-update horizon)
Batch size	16 episodes / update
Gradient clip	max-norm 1.0
Max updates / patience	4 000 / 2 000
Episode cap	400 steps
Train maze size	7 × 7 only
Train bank	500 mazes / seed (seeds ≥ 10 000)
Hidden size	256
Design	8 seeds × 14 conditions = 112 agents

3.6 Evaluation Protocol

Trained policies are evaluated zero-shot by greedy rollout on fresh, held-out mazes at sizes 5, 7, 9, 11, and 13, each with 200 mazes from a reserved high-numbered seed band. Sizes 5–9 bracket the training size; 11 and 13 are strict extrapolation. All conditions and seeds see identical maze banks, so every comparison is exactly paired.

For solved mazes we report the ratio of the agent's step count to the BFS shortest path. The reference is computed on the same mazes the agent solved so that a condition solving fewer mazes is not flattered by dropping its hard cases.

To localise where any turnover effect lives, we ablate gating (forcing all alive gates to 1) and homeostasis at test time in a 2 × 2 design, paired per checkpoint.

We treat the trained agent, not the individual maze, as the unit of analysis. Every condition is evaluated on the same held-out mazes,

so we compare each CTT condition against a baseline seed by seed and analyse the paired differences. These differences are tested with seed-blocked models, which remove the variation between seeds and make no assumption about a random-effects distribution. The primary test is a two-way ANOVA over conditions and seeds. To it we add a more sensitive ordered-trend test: a within-seed regression that asks whether generalisation changes smoothly with turnover magnitude and with regime. Because this test looks for a single direction, it uses only one degree of freedom and so has more power than the ANOVA to detect a graded effect. Solve rate is an overdispersed proportion—its variation between seeds is far larger than the sampling noise within one agent (a dispersion factor of about twenty). Treating each maze as an independent observation would overstate significance; the trained agent (one condition–seed pair) therefore remains the unit of replication. We confirm the headline results on this scale with a binomial generalised linear model whose standard errors are cluster-robust (clustered on the seed). Family-wise error is controlled with the Holm correction [8], and the key contrasts are cross-checked with sign-flip permutation tests. Finally, we report a per-condition analysis alongside the per-seed one and raise the resulting pseudoreplication concern ourselves (Section 4.5).

3.7 Implementation and Reproducibility

The system is implemented in Python and PyTorch. The turnover engine is reused directly from the source implementation, with a small adapter exposing the (kill, birth) configuration; reusing rather than reimplementing it decreases the possibility for implementation errors and keeps the mechanism similar to the source. A unit-test suite asserts that the mechanism does what is specified: the kill/birth arithmetic and the alive-count trajectory, the top-k and newborn protections and the minimum-alive floor, clone-mutate donor selection, the linear gate ramp, parameter-count invariance across all conditions, and checkpoint round-trip including restoration of the homeostasis state. All maze banks are seed-deterministic and the training pipeline is resumable, so the full 112-agent study reproduces from seeds alone.

3.8 Ethics

This study used only procedurally generated maze environments and simulated agents; no human or animal data were collected, so no ethical clearance was required.

4 Results and Discussion

We organise the findings as follows. Sections 4.1–4.2 answer the research question directly, establishing that the ratio does not affect scale generalisation and that this is a bounded null. Section 4.3 shows that the apparent low-power effect was sampling noise and that the null survives the statistically correct model; Sections 4.4–4.7 report the two effects that are real, though orthogonal to the ratio itself; Section 4.8 draws these together into the study's main contributions. Throughout, the unit of replication is the trained agent (a condition–seed pair, $n = 8$), because between-seed training variance dominates the within-model sampling noise roughly twentyfold (overdispersion $\varphi \approx 20$). Table 3 collects the headline tests.

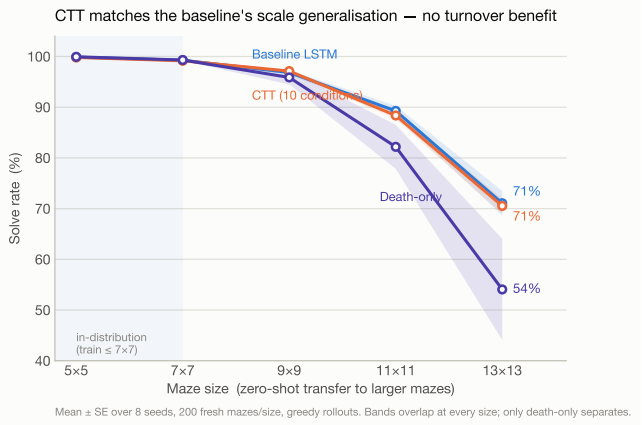


Figure 3: Scale generalisation. Mean solve rate versus maze size for the pooled CTT conditions (orange), the vanilla/dropout/zoneout baselines (blue), and the death-only ablation (purple), with shaded bands showing ± standard error over 8 seeds and 200 held-out mazes per size. The shaded background marks the 5×5–7×7 training range; 9×9 and above are zero-shot extrapolation. CTT tracks the baselines almost exactly at every size, including the two extrapolation sizes that answer the research question; only the death-only ablation, which never rebuilds killed capacity, departs.

4.1 The omnibus test finds no condition effect

Sizes 5–9 are non-discriminating: every condition solves near ceiling ($\geq 97\%$). Only 11×11 and 13×13 separate anything (Figure 3). At 13×13 the regime-pooled solve rates cluster tightly — baseline 71.1%, net-shrinking 72.4%, net-growth 71.6%, balanced 68.3% — with the death-only ablation (54.1%) the sole departure. A two-way ANOVA (condition × seed, blocking on seed) over the thirteen conditions excluding the death-only ablation finds no condition effect, $F(12, 84) = 0.94$, $p = 0.51$; the between-condition spread is no larger than the residual seed-to-seed noise. Including death-only inflates this only to $F(13, 91) = 1.55$, $p = 0.12$.

The omnibus test finds no condition effect at either extrapolation size. This is the primary analysis, and it directly answers the research question: no, the kill/birth ratio has no effect on generalisation performance.

4.2 No individual kill/birth-ratio contrast reaches significance

The predicted direction that balanced turnover generalises better than unbalanced is unsupported and is, if anything, slightly reversed: balanced − unbalanced = −3.6 pp (95% CI $[-12.3, +5.0]$), $t(7) = -1.00$, $p = 0.35$. Every kill/birth-ratio contrast overlaps the baseline band and none survives Holm correction (Figure 4). Crucially, this is a *bounded* null rather than a mere non-detection: with eight seeds we can exclude any advantage of one turnover regime over another larger than roughly a dozen percentage points of solve rate. Turnover as a whole is neutral (CTT − baseline = −2.0 pp, $p = 0.29$).

Table 3: Headline statistical tests. Estimates are in percentage points (pp) of solve rate unless noted; the primary tests are at the 13×13 extrapolation size. CIs are 95%. "robust" denotes the cluster-robust binomial refit that accounts for overdispersion.

Test	Estimate	95% CI	Statistic, p
Omnibus, condition×seed ANOVA (excl. death-only)	—	—	$F(12, 84) = 0.94,\ p = 0.51$
Predicted direction: balanced − unbalanced	−3.6	$[-12.3, +5.0]$	$t(7) = -1.00,\ p = 0.35$
Ordered-trend, regime (post-hoc, robust)	—	—	$p = 0.27$ (naïve 0.05)
Death-only − baseline	−17.0	$[-39.4, +5.4]$	$t(7) = -1.79,\ p = 0.12$
Suboptimality gap, per seed	+0.078	—	$t = 2.1,\ p < 0.05$

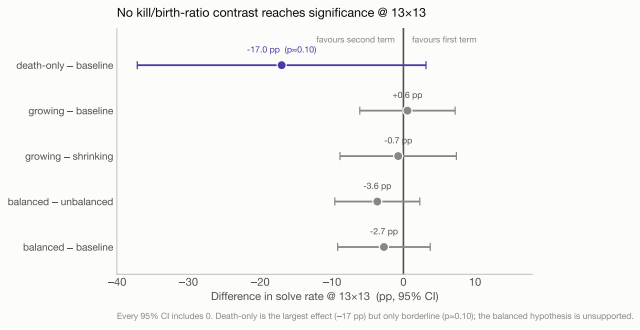


Figure 4: Per-contrast solve-rate differences from the baseline band at 13×13 with 95% confidence intervals.

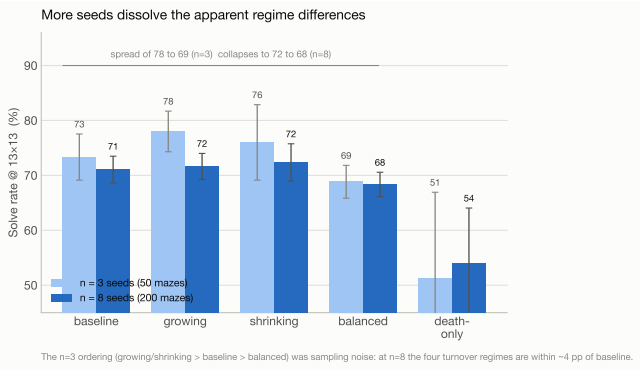


Figure 5: Solve rate at 13×13 for each turnover regime (baseline, net-growth, net-shrinking, balanced, death-only), computed first from the initial three seeds (light bars) and again after five more seeds were added (dark bars, $n = 8$); error bars show ± standard error.

This reinforces the same answer to the research question at the level of individual contrasts, not just the pooled omnibus test: because every contrast overlaps the baseline band and the predicted balanced-versus-unbalanced direction is if anything reversed in sign, the null is not merely undetected but bounded away from any effect larger than about a dozen percentage points.

4.3 The apparent effect was sampling noise, and the null is robust

The regime differences that first motivated the study turned out to be an artefact of low statistical power. With three seeds, the three turnover regimes spanned 7.8 pp and net-growth looked best; with eight seeds, that spread shrank to 4.0 pp and the ranking reshuffled (Figure 5). We then applied the ordered-trend test, which is a within-seed regression that tests for a graded effect of turnover magnitude and regime, and is more powerful than the omnibus ANOVA because it looks for a single direction. Only one of its results fell below the 0.05 threshold: the regime slope at 13×13 ($p = 0.047$). This lone result does not survive closer inspection. It disappears when magnitude and regime are fitted together ($p = 0.30$); when the two outlier conditions are removed (the two axes then jointly explain under 1% of the variance, $R^2 = 0.005$); and when the five maze sizes are corrected for multiple testing (smallest adjusted $p = 0.20$). As a final check we refit the trend on the correct scale. Because solve rate is overdispersed, we used a binomial model with cluster-robust standard errors; its coefficients then give $p = 0.27$ and 0.28, confirming the null. A naive model that treated each maze as independent would instead have reported $p < 0.001$—a false positive produced entirely by ignoring the variation between seeds. Since the ordered-trend test was chosen after seeing the data, we report it only as a robustness check, not as evidence of an effect.

This closes off the one result that had looked like support for an effect and reinforces that the ratio has no detectable effect on generalisation once seed count and overdispersion are properly accounted for.

4.4 Removing the birth step is costly: the death-only floor

The death-only ablation (kills with no replacement) is the one condition that visibly departs from the pack. It degrades fastest with maze size and is worst on every axis, reaching −17.0 pp versus the baselines at 13×13 ($t(7) = -1.79$). We read this as a *floor effect at zero rebuilding* rather than a monotone benefit of birth: kill-without-replace permanently removes plastic capacity with no route back, but among the conditions that *do* rebuild, the smooth birth-fraction gradient is absent (it weakens from $t = 1.50$ at $n = 3$ to $t = 0.81$ at $n = 8$). Rebuilding matters; the *amount* of rebuilding, beyond having any, does not.

This is not evidence that the ratio matters — it is a floor effect at zero rebuilding, and the research question concerns the ratio *among* conditions that do rebuild, where no gradient is present.

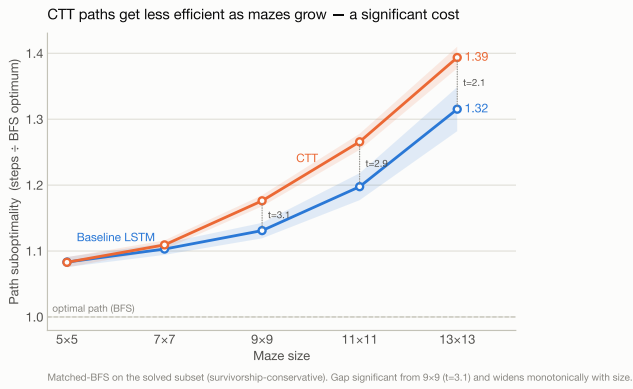


Figure 6: Path suboptimality (agent steps divided by the BFS-optimal path length, matched on the maze subset each condition actually solved) versus maze size, for the baselines (blue) and the pooled CTT conditions excluding death-only (orange), mean ± standard error over 8 seeds.

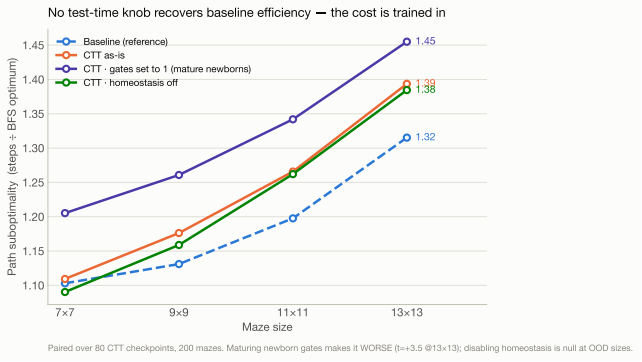


Figure 7: Test-time ablation of the CTT mechanism on already-trained checkpoints: the dashed blue line is the baseline reference; orange is CTT evaluated normally; purple forces every newborn's gate fully open, skipping the gradual maturation ramp; green disables the homeostatic rescaling step.

4.5 Turnover imposes a trained-in path-efficiency cost

CTT agents (excluding death-only) reach the goal as often as the baselines but along measurably longer paths, and the gap widens monotonically with maze size: the suboptimality (agent steps ÷ BFS optimum) exceeds the baselines by +0.045, +0.068, and +0.078 at 9×9, 11×11, and 13×13, with per-seed $t = 3.1, 2.7, 2.8$ ($p = 0.016, 0.030, 0.028$; Figure 6). The comparison is conservative: CTT solves marginally fewer mazes, so survivorship would bias the gap *downward*.

This cost is orthogonal to the research question, which concerns solve-rate generalisation rather than path efficiency, but it shows the ratio is not simply inert — it has a measurable effect, just not on the outcome the research question asks about.

4.6 The cost is baked into the trained weights, not a removable artefact

We localise the cost with a 2×2 test-time ablation of the gating and homeostasis mechanisms (Figure 7): no inference-time configuration recovers baseline efficiency — forcing newborn gates fully open makes suboptimality *worse* (+0.061 at 13×13, $t = +3.5$) and homeostasis-off is null at the extrapolation sizes, so the cost is baked into the trained weights, not a removable artefact.

Localising the cost to the trained weights rules out a simple test-time explanation. Solve-rate generalisation is still unaffected by the ratio, as Section 4.1 established, but it strengthens the case that the path-efficiency cost is a genuine, trained-in property of turnover rather than a runtime artefact.

4.7 The cost belongs to turnover, not the plastic layer

To determine whether this cost is a property of the turnover *process* or merely of the added plastic layer, we trained a static-plastic control: an architecturally identical plastic layer with the turnover cycle disabled, so that all 256 active units persist and train by gradient

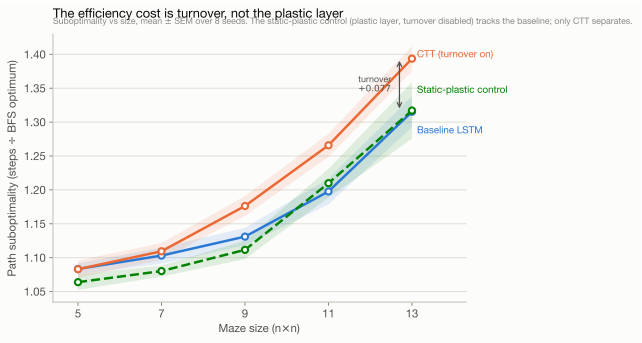


Figure 8: The efficiency cost is turnover, not the plastic layer. Path suboptimality versus maze size for the baselines, the static-plastic control (plastic layer, turnover disabled), and CTT (mean ± SEM over 8 seeds). The control tracks the baselines; only CTT separates, and the gap it opens is the turnover component.

descent alone (no unit is ever scored, killed, or reborn). Evaluated on the same held-out mazes, the control yields an unambiguous decomposition (Figure 8). The plastic layer alone imposes no detectable efficiency cost: the control tracks the baselines at every size (layer component −0.020, +0.012, +0.002 at 9×9, 11×11, 13×13; all non-significant, $p \geq 0.28$). The turnover process accounts for essentially the entire gap (turnover component +0.065, +0.056, +0.077; significant at 9×9, $p = 0.003$, and 11×11, $p = 0.048$, and of equal magnitude at 13×13, $p = 0.096$). Solve rate is unaffected throughout (static-plastic versus baseline −4.3 pp, $p = 0.52$; CTT versus static-plastic +3.8 pp, $p = 0.53$). The trained-in path-efficiency cost is therefore attributable to turnover *per se*, not to the presence of the plastic layer.

We are candid about the analysis unit here. The *per-seed* suboptimality gap is significant and survives Holm correction within its

family at all sizes; the *per-condition* test, with three baseline architectures, is not ($p = 0.08$–0.21). This is a pseudoreplication concern that we raise. We therefore lean on the monotone consistency of the trend across sizes (per-condition $t = 2.79 \rightarrow 2.27 \rightarrow 1.64$, all positive) rather than on any single p-value.

Attributing the cost to turnover rather than the plastic layer confirms that the kill/birth ratio, not the architecture change it requires, is the operative variable — consistent with the research question's framing even though the effect it produces is on path efficiency rather than generalisation.

4.8 Contributions

Our null does not contradict the source work that introduced CTT. That work validated *damage tolerance* on supervised sequence-memory tasks; we tested *scale* generalisation in reinforcement learning — a different axis. Its benefit is explicitly regime-conditional, largest where memory demand is highest and absent where all methods already saturate. Our maze task behaves like that saturated case at sizes 5–9, and only 13×13 separates anything; the reconciliation is therefore not that turnover fails, but that our task does not exercise the axis on which turnover helps. It is also worth noting that our turnover rates (5–30% of the live population per event) are far more aggressive than the canonical symmetric 2%, and that the condition closest to the source configuration is statistically indistinguishable from the baselines.

Across the scale sweep, the only manipulation that moves performance is the removal of capacity (death-only). No regime that rebuilds capacity improves generalisation over the baselines, and no gradient of the kill/birth ratio predicts robustness. The reduced path efficiency that turnover *does* impose is a training-time regularisation effect localised to the weights by our ablation. These support a single reading: in this reinforcement-learning regime the kill/birth ratio behaves as a capacity knob, and CTT's benefits, consistent with its damage-tolerance, are about robustness rather than generalisation, which supports prior work surrounding CTT.

This picture is consistent with the neurogenesis-as-regulariser literature: Tran et al. [16] found that a neurogenesis-like turnover process matches conventional regularisers (dropout, weight decay) on generalisation rather than beating them, which is exactly what we observe when turnover is added to a recurrent policy. Chambers and Conroy [1] modelled turnover rates as something that should be *geared to novelty*; our fixed-schedule, fixed-task setting presents little novelty variation, which, extended to our setting, would predict no advantage for any particular fixed ratio, aligning with our findings.

Beyond the CTT-specific findings, the study is a cautionary case for empirical deep-RL practice. The $n = 3 \rightarrow n = 8$ collapse shows how readily an under-powered seed count manufactures a regime "effect"; the overdispersion diagnosis ($\varphi \approx 20$) shows why maze-level or episode-level analyses are anti-conservative and why the trained agent must be the unit of replication; and reporting bounded nulls with confidence intervals, rather than a bare "no effect", is both more honest and more informative.

Drawing the section together, this study's primary contribution — beyond the headline null that the kill/birth ratio has no detectable effect on generalisation — is twofold: First, it demonstrates empirically that a structural-plasticity mechanism modelled on adult neurogenesis can behave as a *robustness* mechanism without being a *generalisation* mechanism, even when its rebuild/discard balance is swept deliberately and widely: the two properties are dissociable, and a mechanism validated on one should not be assumed to transfer to the other. No prior work has tested both properties within the same design for a turnover mechanism, but we can now say with certainty that one does not imply the other. Second, it demonstrates a concrete case in which a biologically-motivated intervention imposes a measurable, trained-in behavioural cost (reduced path efficiency) with no accompanying benefit on the metric it was hypothesised to improve — a reminder that fidelity to a biological mechanism does not by itself guarantee a computational advantage if the task and environment are not correctly suited to the system.

5 Conclusions

We isolated the kill/birth ratio in Complementary Turnover Training and asked whether it affects an LSTM navigation policy's ability to generalise to *larger* mazes than any seen in training. The answer is no, and, importantly, the null is bounded: no turnover regime beats the regularised baselines, and any balanced-versus-unbalanced regime difference is confined to under about a dozen percentage points of solve rate. Two findings survive scrutiny: removing rebuilding is costly, but as a floor effect at zero rebuilding rather than a gradient (the death-only ablation); and turnover imposes a path-efficiency cost that widens with maze size and that a test-time ablation localises to the trained weights. In this reinforcement-learning regime the kill/birth ratio is a capacity knob computation as a whole, since it separates what a neurogenesis-like mechanism can be credited with (robustness) from what it cannot (better generalisation), purely by tuning how aggressively capacity is rebuilt.

5.1 Future Work

Two limitations point most directly to future work. The task family used here places only a light demand on memory, with ceiling effects below 11×11 that compress the usable signal into the two largest sizes; repeating this sweep on a genuinely memory-intensive task such as copy-800, used in the source work's lesion-robustness study, would test whether the null found here is a property of the kill/birth ratio itself or of a task that does not exercise the axis on which CTT is expected to help. Separately, three of CTT's four design dimensions — the scoring rule, the rebuilding strategy, and the stabilisation machinery — were inherited unchanged from the source configuration, so these conclusions concern the ratio dimension only; sweeping those dimensions, and in particular testing adaptive, novelty-gated scheduling in the manner proposed by Chambers and Conroy [1], would establish whether the null generalises across CTT's design space or is specific to a fixed schedule.

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A AI Usage Statement

AI was used for clarifying the discussion and conclusions sections as well as generating ideas for subheadings of sections. It was also used to proofread for the general flow of the report, as well as ensure that my abstract was a strong enough representation of my honours project as a whole, since it is the first paragraph that sets the tone for the entire paper. However, no writing was copy-and-pasted and all body paragraphs of this paper are my own writing. All em-dashes used in this document are intentional, chosen by me for correct grammar and clarity.

B Supplementary Information

This appendix provides the full statistical detail underlying Section 4. All values are computed from the raw evaluation log; a workbook of every test (estimates, 95% CIs, raw and Holm-adjusted p-values, and the cluster-robust refit) is included in repository.

B.1 Ordered-trend slopes by maze size

Within-seed regression of solve rate on turnover magnitude ($\log_{10}(\text{kill}+\text{birth})$) and regime ($(\text{birth}-\text{kill})/(\text{birth}+\text{kill})$), slopes in pp per axis unit. Nothing survives Holm across the five sizes (minimum corrected $p = 0.20$).

Size	Magnitude slope	p	Regime slope	p
5×5	−0.08	0.22	−0.02	0.75
7×7	−0.05	0.84	−0.06	0.80
9×9	+1.10	0.10	+0.54	0.36
11×11	+3.05	0.15	+3.73	0.05
13×13	+8.81	0.06	+8.40	0.05

Table 4: Ordered-trend slopes by size (univariate, within-seed).

B.2 Correct-model check (binomial, cluster-robust)

Solve rate is an overdispersed proportion ($\varphi = 17$–25); a naïve maze-level binomial is anti-conservative. The cluster-robust refit confirms every seed-level null.

Term	Model	Naïve p	Robust p
magnitude (11 CTT)	GLM, cluster by seed	0.0004	0.28
regime (11 CTT)	GLM, cluster by seed	0.00002	0.27

Table 5: Naïve versus cluster-robust binomial p-values.